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2026

MACHINE LEARNING

3/2/2026

PHASE 1: Foundations & Light ML Exposure

Lane 1 — Mathematics Core

- ❿ Linear Algebra (primary focus)
- ❿ Selected chapters from: Mathematics for Machine Learning
- ❿ Focus areas:
    - ❿ Vector spaces
    - ❿ Matrix multiplication
    - ❿ Eigenvalues & eigenvectors
    - ❿ Gradients

Lane 2 — Python Engineering

- ❿ Core Python
- ❿ Numpy (numerical computing)
- ❿ Basic OOP (classes, inheritance)
- ❿ Data structures (lists, dicts, sets, tuples)
- ❿ References:
    - ❿ Official Python documentation
    - ❿ NumPy tutorials

Lane 3 — Light ML Exposure

- ❿ Linear regression
- ❿ Logistic regression
- ❿ Overfitting & underfitting concepts
- ❿ Tools: scikit-learn
- ❿ Goal: Understand model behavior, not full derivations

Lane 4 — Data Handling & Visualization

- ❿ Pandas (data manipulation)
- ❿ Matplotlib & Seaborn (visualization)
- ❿ Integrate visualizations with ML models to build intuition

PHASE 2: Intermediate ML & Mathematics

Lane 1 — Calculus & Optimization

- ❿ Multivariable calculus
- ❿ Chain rule
- ❿ Gradient descent
- ❿ Convex optimization basics
- ❿ Continue Deisenroth

Lane 2 — Probability & Statistics

- ❿ Discrete probability
- ❿ Random variables
- ❿ Expectation & variance
- ❿ Gaussian distribution
- ❿ Gradually: All of Statistics

Lane 3 — ML Foundations (Serious Level)

- ❿ Derive linear regression & logistic regression
- ❿ Regularization (L1, L2)
- ❿ Bias–variance tradeoff
- ❿ Use: Hands-On Machine Learning (Géron) — critically, not just coding

Lane 4 — Practical Probability Experiments in Python

- ❿ Dice rolls, random sampling, Gaussian simulations
- ❿ Integrate with visualization & small regression experiments

PHASE 3: Advanced ML & Deep Learning

Lane 1 — Advanced ML Theory

- ❿ SVM derivation & kernel trick
- ❿ EM algorithm
- ❿ PCA with full linear algebra reasoning

Lane 2 — Deep Learning

- ❿ PyTorch basics
- ❿ Backpropagation derivation

- ❿ Neural networks from scratch
- ❿ CNN fundamentals
- ❿ Eventually: Deep Learning

Lane 3 — Engineering Practice

- ❿ Git version control
- ❿ Linux basics
- ❿ Writing clean, modular code
- ❿ Experiment tracking & reproducibility

PHASE X: Pre-Creation

Lane 0 — Engineering Best Practices

- ❿ Unit testing
- ❿ Logging
- ❿ Virtual environments & dependency management

Lane 1 — Research Exposure

- ❿ Study research papers: GPT-2, GPT-3, LLaMA
- ❿ Analyze architectures, scaling, and training strategies

Lane 2 — Small Domain Tools

- ❿ Build small domain-specific AI tools
- ❿ Experiment with data pipelines & simple model deployment

Lane 3 — Open Source Contribution

- ❿ Contribute to ML projects to understand large-scale training practices

PHASE Q: Creation Phase

Lane 1 — Sequence Models

- ❿ Small sequence models or LSTM on toy datasets

Lane 2 — Transformers

- ❿ Implement transformer models using PyTorch / Hugging Face
- ❿ Start with small-scale projects, gradually increase complexity

Lane 3 — Research Integration

- ❿ Read and analyze research papers of GPT-2, GPT-3, LLaMA

- ❿ Apply insights to your experiments

Lane 4 — Domain-Specific AI Tools

- ❿ Build full AI tools in your target domain using learned concepts

Lane 5 — Open Source Collaboration

- ❿ Contribute to existing large-scale projects
- ❿ Learn scaling, versioning, deployment practices